RL-GAN Multi-Image Ablation Evaluation Report

> **Architecture**: Places2-pretrained DeepFillV2 backbone + COCO-val2017-trained adapters + PPO-trained RL controller.

Ablation Stage Key

Stage	Description	Scientifically Valid?
A	DeepFillV2 backbone only (no adapters, no RL)	Yes
B	Fixed adapter 0 (Global) — no RL selection	Yes
C	PPO-trained adapter selection — pure inference	Yes
D	Stage C + 400-step GT-supervised AdamW refinement	Uses ground truth

Quantitative Ablation Table

#	Scene	RL Action	A: Backbone	B: Fixed Adapter	C: PPO-only	D: PPO+Refine	Delta C vs A	Delta D vs A
1	Alpine Lake & Snowy Mountains	Action 2: Boundary Refinement	34.45 dB	35.21 dB	**32.25 dB**	37.23 dB	**-2.21 dB**	+2.78 dB
2	Lush Alpine Meadow & Distant Peaks	Action 0: Global Completion	33.72 dB	34.54 dB	**34.54 dB**	39.65 dB	**+0.82 dB**	+5.94 dB
3	Sunset Mountain Horizon & Dusk Sky	Action 2: Boundary Refinement	40.58 dB	40.00 dB	**36.80 dB**	43.19 dB	**-3.78 dB**	+2.61 dB

Detailed Scene Analysis

Test 1: Alpine Lake & Snowy Mountains

- **Scene**: Turquoise mountain lake with crystal water reflections and forested ridges.
- **Source**: `beautiful-lake-mountains_395237-44.avif`
- **PPO**: Trained PPO agent
- **Selected Action**: Action 2: Boundary Refinement

Stage	PSNR	SSIM	vs Backbone
A: Backbone only	34.45 dB	0.9768	baseline

Stage	PSNR	SSIM	vs Backbone
B: Fixed adapter	35.21 dB	0.9784	+0.75 dB
C: PPO-only (pure inference)	32.25 dB	0.9730	**-2.21 dB**
D: PPO+Refinement (GT-supervised)	37.23 dB	0.9828	+2.78 dB

- **Output**: `demo_image_1_alpine_lake.png`

Test 2: Lush Alpine Meadow & Distant Peaks

- **Scene**: Vibrant green grassy slopes, scattered fir trees, and fluffy summer clouds.
- **Source**: `beautiful-natural-image-1844362_1280.jpg`
- **PPO**: Trained PPO agent
- **Selected Action**: Action 0: Global Completion

Stage	PSNR	SSIM	vs Backbone
A: Backbone only	33.72 dB	0.9703	baseline
B: Fixed adapter	34.54 dB	0.9734	+0.82 dB
C: PPO-only (pure inference)	34.54 dB	0.9734	**+0.82 dB**
D: PPO+Refinement (GT-supervised)	39.65 dB	0.9872	+5.94 dB

- **Output**: `demo_image_2_lush_meadow.png`

Test 3: Sunset Mountain Horizon & Dusk Sky

- **Scene**: Jagged snow-covered crests under pink/purple twilight horizon gradients.
- **Source**: `photo-masthead-375-BoK_p8LG.webp`
- **PPO**: Trained PPO agent
- **Selected Action**: Action 2: Boundary Refinement

Stage	PSNR	SSIM	vs Backbone
A: Backbone only	40.58 dB	0.9908	baseline
B: Fixed adapter	40.00 dB	0.9900	-0.58 dB
C: PPO-only (pure inference)	36.80 dB	0.9834	**-3.78 dB**
D: PPO+Refinement (GT-supervised)	43.19 dB	0.9946	+2.61 dB

- **Output**: `demo_image_3_sunset_peaks.png`

Methodology Notes

- **Stage C (PPO-only)** is the correct measurement of the RL contribution. Single forward pass with PPO-selected adapter — no optimization, no ground truth used.

- **Stage D (PPO + Refinement)** applies 400 AdamW steps with ground-truth in the loss. This is test-time supervised fine-tuning — not a valid generalization metric.
- **Backbone**: DeepFillV2 pretrained on Places2 (frozen during adapter and RL training).
- **Adapters**: Trained 3 epochs on COCO val2017 (5,000 images, synthetic masks).
- **PPO**: Trained on COCO val2017 via stable-baselines3 PPO.